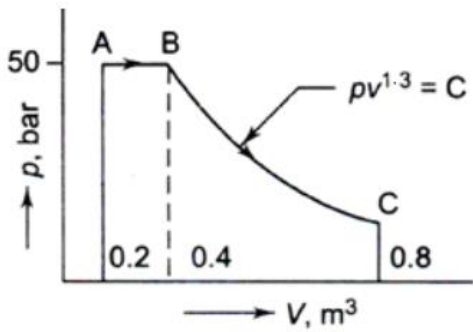
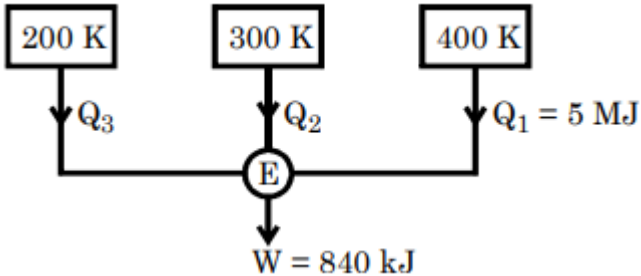


DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Summer Examination – 2023 Course: B. Tech. Branch : Mechanical Engineering Semester :III Subject Code & Name: Thermodynamics (BTMC303) Max Marks: 60 Date: 14-08-2023 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Use of steam table is allowed 5. Assume suitable data wherever necessary and mention it clearly.			
		(Level/ CO)	Mar ks
Q. 1	Solve Any Two of the following.		12
A)	Define the thermodynamic system. Classify it with brief explanation and example of each.	L2 - Understand CO1	6
B)	Define thermodynamic equilibrium and classify it with brief explanation of each	L2 - Understand CO1	6
C)	Determine the total work done by a gas system following an expansion process as shown in Figure. 	L3 -Apply CO1	6
Q.2	Solve Any Two of the following.		12
A)	A mass of 8 kg gas expands within a flexible container so that the p–V relationship is of the form $pV^{1.2} = \text{constant}$. The initial pressure is 1000 kPa and the initial volume is 1 m³ . The final pressure is 5 kPa . If specific internal energy of the gas decreases by 40 kJ/kg , find the heat transfer in magnitude and direction.	L3 -Apply CO2	6
B)	A reciprocating air compressor takes in 2 m³/min at 0.11 MPa, 20°C which it delivers at 1.5 MPa, 111°C to an aftercooler where the air is cooled at constant pressure to 25°C . The power absorbed by the compressor is 4.15 kW . Determine the heat transfer in (a) The compressor (b) The cooler. State your assumptions.	L3 -Apply CO2	6

C)	Explain different forms of energy and write down the equation of each.	L2 - Understand CO2	6
Q. 3	Solve Any Two of the following.		12
A)	Two reversible heat engines A and B are arranged in series, A rejecting heat directly to B. Engine A receives 200 kJ at a temperature of 421°C from a hot source, while engine B is in communication with a cold sink at a temperature of 4.4°C . If the work output of A is twice that of B, find (a) The intermediate temperature between A and B (b) The efficiency of each engine (c) The heat rejected to the cold sink	L3 -Apply CO3	6
B)	A reversible engine, as shown in Figure during a cycle of operations draws 5 MJ from the 400 K reservoir and does 840 kJ of work. Find the amount and direction of heat interaction with other reservoirs. 	L3 -Apply CO3	6
C)	State and explain the following statements of second law of thermodynamics 1. Clausius Statement of the Second Law of thermodynamics. 2. Kelvin–Planck Statement of the Second Law of thermodynamics.	L2 - Understand CO3	6
Q.4	Solve Any Two of the following.		12
A)	Air contained in a cylinder fitted with a piston is compressed reversibly according to the law $pV^{1.25} = \text{const}$. The mass of air in the cylinder is 0.1 kg . The initial pressure is 100 kPa and the initial temperature 20°C . The final volume is 1/ 8 of the initial volume. Determine the work and the heat transfer.	L3 -Apply CO4	6
B)	A closed system allows nitrogen to expand reversibly from a volume of 0.25 m³ to 0.75 m³ along the path $pV^{1.32} = \text{const}$. The original pressure of the gas is 250 kPa and its initial temperature is 100°C. (a) Draw the p-v and T-s diagrams. (b) What are the final temperature and the final pressure of the gas? (c) How much work is done and how much heat is transferred? (d) What is the Entropy change of nitrogen?	L3 -Apply CO4	6
C)	State the following laws related to ideal gases and write the their equations 1. Boyle's Law 2. Charles' Law 3. Gay Lussac's Law 4. Avogadro's Law	L2 - Understand CO4	6

Q. 5	Solve Any Two of the following.		12
A)	Explain the phase change process of the pure substance and plot the same on T-s Plot by assume the initial state of the substance to be solid ice and it is heated at constant pressure.	L2 - Understand CO5	6
B)	Define the following terms related to pure substances (only definitions) 1. Saturated Liquid. 2. Saturated Vapor States 3. Saturated Liquid–Vapor Mixture 4. Superheated Vapor 5. Compressed Liquid 6.Dryness Fraction	L2 - Understand CO5	6
C)	Find the enthalpy and entropy of steam when the pressure is 2 MPa and the specific volume is 0.09 m³/kg .	L3 -Apply CO5	6
	*** End ***		

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